Coordinate Geometry & Transformations Error Analysis

TEACHER KEY Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Distance formula

ORIGINAL PROBLEM

Find the distance between (1, 2) and (4, 6).

INCORRECT WORKED SOLUTION

$d = |4 - 1| + |6 - 2|$

$d = 3 + 4 = 7$

A. EXPLAIN THE MISTAKE (1 POINT)

The distance formula SQUARES the differences, adds them, and takes a square root: $d = \sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{25} = 5$.

B. SHOW CORRECT WORK (1 POINT)

$d = \sqrt{(4 - 1)^2 + (6 - 2)^2}$

$d = \sqrt{9 + 16}$

$d = \sqrt{25} = 5$

2. Midpoint formula

ORIGINAL PROBLEM

Find the midpoint of (−2, 6) and (4, −2).

INCORRECT WORKED SOLUTION

$M = \left(\frac{4 - (-2)}{2}, \frac{6 - (-2)}{2}\right)$

$M = (3, 4)$

A. EXPLAIN THE MISTAKE (1 POINT)

The midpoint formula ADDS the coordinates: $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$. The wrong work subtracted them.

B. SHOW CORRECT WORK (1 POINT)

$M = \left(\frac{-2 + 4}{2}, \frac{6 + (-2)}{2}\right)$

$M = \left(\frac{2}{2}, \frac{4}{2}\right)$

$M = (1, 2)$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Coordinate Geometry & Transformations Error Analysis

TEACHER KEY Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

3. Classifying quadrilaterals

ORIGINAL PROBLEM

Is ABCD with A(0, 0), B(4, 0), C(3, 3), D(1, 3) a parallelogram?

INCORRECT WORKED SOLUTION

Slope of AB = 0 and slope of DC = 0.

AB ∥ DC.

So ABCD is a parallelogram.

A. EXPLAIN THE MISTAKE (1 POINT)

A parallelogram needs BOTH pairs of opposite sides parallel. Slope of BC = −3 but slope of AD = 3, so BC is not parallel to AD.

B. SHOW CORRECT WORK (1 POINT)

AB ∥ DC (both slopes are 0)

Slope of BC = $\frac{3-0}{3-4} = -3$

Slope of AD = $\frac{3-0}{1-0} = 3$

BC is not parallel to AD, so ABCD is a trapezoid, not a parallelogram.

4. Reflection

ORIGINAL PROBLEM

Reflect (3, −2) across the x-axis.

INCORRECT WORKED SOLUTION

Rule: (x, y) → (−x, y)

(3, −2) → (−3, −2)

A. EXPLAIN THE MISTAKE (1 POINT)

Reflecting across the x-axis changes the sign of y: (x, y) → (x, −y). The wrong rule is a reflection across the y-axis.

B. SHOW CORRECT WORK (1 POINT)

Rule: (x, y) → (x, −y)

(3, −2) → (3, 2)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Coordinate Geometry & Transformations Error Analysis

TEACHER KEY Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

5. Reflection over y = x

ORIGINAL PROBLEM

Reflect (2, 5) across the line y = x.

INCORRECT WORKED SOLUTION

Rule: (x, y) → (−y, −x)

(2, 5) → (−5, −2)

A. EXPLAIN THE MISTAKE (1 POINT)

Reflecting across y = x SWAPS the coordinates: (x, y) → (y, x). The wrong rule belongs to the line y = −x.

B. SHOW CORRECT WORK (1 POINT)

Rule: (x, y) → (y, x)

(2, 5) → (5, 2)

6. Rotation

ORIGINAL PROBLEM

Rotate (4, 1) 90° counterclockwise about the origin.

INCORRECT WORKED SOLUTION

Rule: (x, y) → (y, −x)

(4, 1) → (1, −4)

A. EXPLAIN THE MISTAKE (1 POINT)

A 90° COUNTERCLOCKWISE rotation is (x, y) → (−y, x). The wrong rule rotates clockwise.

B. SHOW CORRECT WORK (1 POINT)

Rule: (x, y) → (−y, x)

(4, 1) → (−1, 4)

Coordinate Geometry & Transformations Error Analysis

TEACHER KEY Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

7. Translation

ORIGINAL PROBLEM

Translate $(-3, 5)$ by the vector $\langle 4, -7 \rangle$.

INCORRECT WORKED SOLUTION

$(-3 - 4, 5 + 7)$

$(-7, 12)$

A. EXPLAIN THE MISTAKE (1 POINT)

ADD the vector's components: $x + 4$ and $y + (-7)$. The wrong work subtracted 4 and added 7.

B. SHOW CORRECT WORK (1 POINT)

$(-3 + 4, 5 + (-7))$

$(1, -2)$

8. Dilation

ORIGINAL PROBLEM

Dilate $(2, 1)$ by a scale factor of 3 about the origin.

INCORRECT WORKED SOLUTION

$(2 + 3, 1 + 3)$

$(5, 4)$

A. EXPLAIN THE MISTAKE (1 POINT)

A dilation MULTIPLIES each coordinate by the scale factor. Adding is a translation.

B. SHOW CORRECT WORK (1 POINT)

$(x, y) \rightarrow (3x, 3y)$

$(2, 1) \rightarrow (6, 3)$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Coordinate Geometry & Transformations Error Analysis

TEACHER KEY Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

9. Dilation (shrinking)

ORIGINAL PROBLEM

Dilate $(6, -4)$ by a scale factor of $\frac{1}{2}$ about the origin.

INCORRECT WORKED SOLUTION

$(x, y) \to (2x, 2y)$

$(6, -4) \to (12, -8)$

A. EXPLAIN THE MISTAKE (1 POINT)

A scale factor of $\frac{1}{2}$ means MULTIPLY by $\frac{1}{2}$ (a shrink), not by 2: $(\frac{1}{2} \cdot 6, \frac{1}{2} \cdot (-4)) = (3, -2)$.

B. SHOW CORRECT WORK (1 POINT)

$(x, y) \to (\frac{1}{2}x, \frac{1}{2}y)$

$(6, -4) \to (3, -2)$

10. Equation of a circle

ORIGINAL PROBLEM

Write the equation of a circle with center $(-2, 3)$ and radius 4.

INCORRECT WORKED SOLUTION

$(x - h)^2 + (y - k)^2 = r^2$

$(x - 2)^2 + (y + 3)^2 = 16$

A. EXPLAIN THE MISTAKE (1 POINT)

The center is $(h, k) = (-2, 3)$, so $x - h = x - (-2) = x + 2$ and $y - k = y - 3$. The signs were reversed.

B. SHOW CORRECT WORK (1 POINT)

$(x - (-2))^2 + (y - 3)^2 = 4^2$

$(x + 2)^2 + (y - 3)^2 = 16$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points